C:\hpc-23-feb-2026\.\25\eg187.cpp

```
#include<oneapi/tbb.h>
#include<vector>
#include<chrono>
#include<iostream>
#include<unordered_set>
using namespace std;
using namespace oneapi::tbb;
int main()
{
unordered_set<thread::id>  thread_ids;
vector<int64_t> v(10000000,10);
auto start=chrono::high_resolution_clock::now();
int64_t total=parallel_reduce(blocked_range<size_t>(0,v.size()),
0LL,
[&](blocked_range<size_t> &range,int64_t acc_sum){
thread_ids.insert(this_thread::get_id());
auto start_at=range.begin();
auto end_at=range.end();
cout<<"s:"<<start_at<<",e:"<<end_at<<endl;
for(auto i=start_at;i!=end_at;++i)
{
acc_sum+=v[i];
}
return acc_sum;
},
[](int p,int q)->int{return p+q;}
);
auto end=chrono::high_resolution_clock::now();
auto duration=chrono::duration_cast<chrono::milliseconds>(end-start);
cout<<duration.count()<<" milliseconds"<<endl;
cout<<"Total "<<total<<endl;
cout<<"Number of threads created : "<<thread_ids.size()<<endl;
return 0;
}
```

C:\hpc-23-feb-2026\.\25\eg188.cpp

```
#include<iostream>
#include<vector>
#include<algorithm>
using namespace std;
int main()
{
vector<int> v={10,20,30,40};
transform(v.begin(),v.end(),v.begin(),[](int value)->int{
return value+5;
});
for(auto x:v) cout<<x<<endl;
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\25\eg189.cpp

```
#include<string>
#include<iostream>
#include<algorithm>
using namespace std;
int main()
{
string s="God is great";
transform(s.begin(),s.end(),s.begin(),[](char c){
return toupper(c);
});
cout<<s<<endl;
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\25\eg190.cpp

```
#include<iostream>
#include<algorithm>
#include<vector>
#include<string>
using namespace std;
int main()
{
vector<int> v={10,20,30,40,50,60};
vector<string> vs(v.size());
transform(v.begin(),v.end(),vs.begin(),[](int num){
return string("Number : ")+to_string(num);
});
for(auto str:vs) cout<<str<<endl;
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\25\eg191.cpp

```
#include<vector>
#include<iostream>
#include<algorithm>
using namespace std;
int main()
{
vector<int> v1={1,2,3,4,5,6,7,8,9,10};
vector<int> v2={10,20,30,40,50,60,70,80,90,100};
vector<int> v3(v1.size());
transform(v1.begin(),v1.end(),v2.begin(),v3.begin(),[](int x,int y){
return x+y;
});
for(auto x:v3) cout<<x<<endl;
return 0;
}
```

C:\hpc-23-feb-2026\.\25\eg192.cpp

```
#include<vector>
#include<numeric>
#include<iostream>
using namespace std;
int main()
{
vector<string> v={"10","20","30","40"};
string str=accumulate(v.begin()+1,v.end(),v[0],[](string a,string b){
return a+string(",")+b;
});
cout<<str<<endl;
return 0;
}
```

C:\hpc-23-feb-2026\.\25\eg193.cpp

```
#include<vector>
#include<numeric>
#include<iostream>
using namespace std;
// this example is not compiling
int main()
{
vector<int> v={10,20,30,40};
string str=accumulate(v.begin()+1,v.end(),to_string(v[0]),[](int a,int b){
return to_string(a)+string(",")+to_string(b);
});
cout<<str<<endl;
return 0;
}
```

Thinking Machines Classroom Codes